

VETO: Supplementary Protocol and Evidence Index

Anonymous CVPR submission

Paper ID DRAFT-S

1. Status

This is an executable-study scaffold, not a completed experimental supplement. Source code, data identities, raw answers, candidate failures, native training receipts and GPU/API costs must be linked before a result is populated.

2. Shared Scoring

Answer extraction is host-owned. Invalid ground truth is a data error; an incorrect or malformed model answer is an unsuccessful trajectory. Pair scores require complete identity-matched predictions on both sides. No invalid or missing sample is silently replaced with a zero or removed from the denominator.

3. Reproducibility

All seeds and condition names are preregistered. The same source family cannot cross data roles. Every real run retains its configuration, source snapshot, model and processor identities, decoder settings, raw responses, and terminal Slurm allocation record. Shared-prefix computations are identified explicitly. The Adam-reset restoration caveat is shared by every trained branch. The public source model is Qwen3.5-4B revision 851bf6e. All conditions start from the same audited BF16 initialization. Native RSFT stores trainable parameters in FP32 and exports BF16 serving weights. Source, derived initialization, native checkpoint and serving identities are recorded separately, with the serving export checked against the native BF16 cast.

4. Meaning of the Paired Criterion

The fractions q_0, q_1, q_2 of pairs with zero, one and two correct answers satisfy $M_C = q_1/2 + q_2$ and $P_C = q_2$, giving $P_C = 2M_C - 1 + q_0$. For an analytical example, an incoming correctness pattern $\{(1, 1), (0, 0)\}$ and a candidate pattern $\{(1, 0), (0, 1)\}$ both have $M_C = 1/2$, but paired correctness falls from $1/2$ to zero. The zero-tolerance marginal gate permits this candidate, whereas the paired gate rejects it. These are constructed correctness patterns, not experimental model scores. The example establishes a possible

distinction only. Actual selection differences, subsequent training outcomes and comparisons to augmentation remain empirical requirements; a nonzero both-incorrect fraction is insufficient evidence.

5. Configuration Calibration

Prompt	Single image	Both images
Direct	83.40	78.91
Structured	88.28	83.59
Brief reasoning	86.33	81.25

Table 1. Complete initial-prompt calibration on256 development pairs. Scores are percentages; these are not VETO gains or test results.

The structured prompt was selected by ordinary development accuracy. Learning-rate calibration is reported separately below. All formal conditions use the selected rate.

Learning rate	Single image	Both images
10^{-7}	86.52	82.81
10^{-6} (selected)	87.50	80.86
10^{-5}	54.69	34.77

Table 2. Complete learning-rate calibration on the same256 development pairs after four native RSFT batches. Scores are percentages. These are configuration measurements, not VETO gains or independent test results.

Ordinary development accuracy determines selection; ties use the smaller rate. Seed42 shares the selected first-stage checkpoint; this calibration is not an additional independent trial.

6. Optimization Archive

Selections: WHALE h3, VETO h3, Marginal gate h3. All three rules select the same harness; this archive supplies no evidence of an independent VETO selection benefit. Unparseable answers on H/C: h0 4/4, h1 4/1, h2 1/2, h3 0/0.

Candidate	H correct	C single	C both
h0	107/128	110/128	49/64
h1	111/128	115/128	54/64
h2	111/128	107/128	48/64
h3	112/128	115/128	53/64

Table 3. Complete fixed-weight optimization archive. These are H/C selection measurements, not independent method results.

These descriptive counts do not identify causal visual improvements.

7. Illustrative Failure and Interpretation

Candidate	Image A	Image B
h0	1900 (correct)	-1850 (correct)
h1	1900 (correct)	-1900 (correct)
h2	2000 (correct)	2000 (wrong)

Table 4. Same-question paired answers for the displayed example.

Exact target values are 1926.7406264900932 and -1926.7406264900932, with the shared 5% relative tolerance. In the displayed h2 response for image B, the model describes East Asia1980 as about700 and Hungary1979 as about2700, then computes2700-700 despite the question asking East Asia1980 minus Hungary1979. Both outputs are2000. The written response reflects an operand-order error; this case does not establish failure to use visual evidence or a latent reasoning mechanism. This example was selected after observing h0/h1/h2 and is not representative-frequency evidence. First lexicographic pair ID among numeric C pairs where h0 and h1 are both correct, h2 is exactly one correct, and both h2 committed answers are parseable numbers. Chosen for illustration after observing h0/h1/h2, not for estimation or optimization. All six displayed trajectories contain no tool calls.

8. Required Evidence

- Main results:18 logical conditions or explicit incomplete/failed records.
- Paired gate versus same-data marginal gate and augmentation.
- External fixed-subset results, including adverse outcomes.
- Source-cluster paired intervals and per-seed raw scores.
- Stage traces, source-grounded success/failure cases and cost curves.

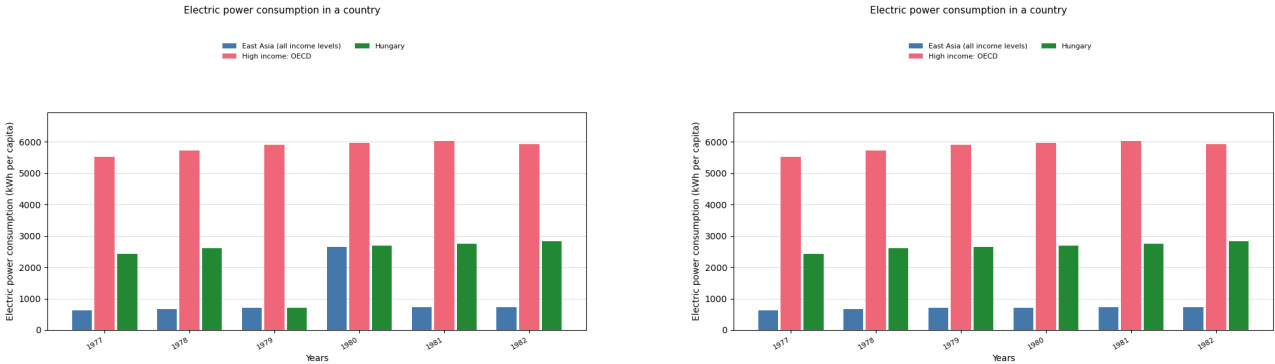


Figure 1. Post-hoc optimization-set illustration, images A (left) and B (right). In the chart, what is the value for "1980" in series "East Asia (all income levels)" minus the value for "1979" in series "Hungary"? Return the numeric value in the plotted axis units, without a unit or percent sign. No inference about which image is the original source is made from display order.